

URL Shortener Web Application - Souryadip Mallick

1. **Project Overview:** The URL Shortener web application is designed to simplify the management and sharing of URLs. Long URLs can be cumbersome to share, copy, and remember. This web application enables users to shorten URLs, copy them with a single click, and maintain a history of their shortened URLs. The application consists of a frontend built using HTML, CSS, and Bootstrap for styling, a backend implemented in Flask (Python), and a database handled via SQLAlchemy ORM with SQLite. The app also includes user authentication to allow personalized URL management. The key objectives of the application are
 - Shortening URLs efficiently.
 - Allowing users to create an account, log in, and manage their URLs.
 - Maintaining a history of URLs shortened by each user along with timestamps.
 - Providing a clean and user-friendly interface for ease of use.
2. **Frontend Design:** The frontend design prioritizes simplicity and clarity while utilizing Bootstrap for responsiveness and style. The application comprises three primary pages:
 - **Home Page:** The home page presents a welcoming message and a concise explanation of the application's purpose. It features two prominent buttons: Login and Sign Up. Both buttons are styled with Bootstrap classes (btn-primary for login and btn-success for signup) and are centered using d-flex justify-content-center. A small footer displays the creator's name. Vanilla CSS was employed minimally for card spacing and custom background colors, while Bootstrap managed the layout and responsiveness.
 - **Login Page:** The login page collects the username and password. Input fields were designed using Bootstrap's form-control class, and the submit button utilizes btn-primary. Flash messages were implemented to provide feedback for incorrect credentials.
 - **Sign Up Page:** The signup page enables users to create an account. It validates username and password criteria: Username: Must be unique (validated against the database) and Password: Must be between 8-16 characters. Flash messages indicate success or failure, and buttons are styled with Bootstrap (btn-success) to maintain consistency.
 - **Dashboard Page:** The dashboard serves as the core functional page where users can input URLs to shorten. Key features include: URL Input Box: For users to paste the URL they wish to shorten, Shorten Button: Initiates the URL shortening process using the pyshorteners Python package, Shortened URL Box: Displays the result and enables users to copy it with a "Copy" button, History Button: Redirects users to their URL history page and a Logout Button: Allows users to terminate their session. The dashboard is designed with a card and a beige background for the page. Rounded corners and subtle shadows enhance its visual appeal.
 - **History Page:** The history page presents a table of all URLs shortened by the logged-in user. Each row displays: Original URL, Shortened URL

and Timestamp of creation. The table is styled using Bootstrap classes such as table-hover, table-responsive, and table-light to ensure readability and aesthetic appeal. The page also includes a Back to Dashboard button for navigation.

3. **Backend Implementation:** The backend is developed using Flask, which handles routing, form submission, session management, and database interactions.

Routing:

- ❖ Home (/) - Displays welcome message with login and signup options.
- ❖ Login (/login) - GET shows form; POST handles authentication.
- ❖ Sign Up (/signup) - GET shows form; POST validates input, creates user, and flashes messages.
- ❖ Dashboard (/dashboard/<username>) - GET shows the dashboard; POST shortens the URL and saves it to history.
- ❖ History (/history/<username>) - GET displays all URLs for the user.
- ❖ Logout (/logout) - Clears session and redirects to home.

Session Management: Flask sessions were introduced to handle user authentication. Upon successful login, session['username'] stores the logged-in user. Access to dashboard and history pages is restricted using session checks. Logging out removes this session key. This ensures security and prevents unauthorized access.

Database: The application uses SQLAlchemy ORM with SQLite. Two models were created:

- ❖ User: Stores username (primary key) and password.
- ❖ URLHistory: Stores username (foreign key), timestamp, original_url, and shortened_url. The timestamp and username together form the composite primary key for URLHistory, ensuring that each shortened URL is uniquely recorded per user.

4. **Bugs and Challenges:** During development, several issues were identified and resolved:

- ❖ OperationalError - No Table: Initially, the database tables were not created. Using db.create_all() under app context solved this issue.
- ❖ Readonly Database Error: Attempting inserts failed when the SQLite file permissions were incorrect. Changing file permissions resolved this.
- ❖ Login Redirection: Using "url_for" with username caused BuildError. Solved by redirecting using an f-string: /dashboard/{username}.
- ❖ Session Handling: Without sessions, users could access other dashboards directly via URL. Implemented Flask sessions to restrict access.
- ❖ Copy Button Issue: Needed JavaScript to copy the shortened URL to clipboard with a feedback animation ("Copied!").
- ❖ History Sorting: Ensured that history is displayed in descending order by timestamp.
- ❖ Button Styling: Some buttons were stretching full width. Fixed using d-inline-block and custom padding for buttons like "Created by" and logout.

5. **Approach and Implementation Notes**: The project followed a modular and iterative approach:

- ❖ **Frontend Setup**: HTML pages with Bootstrap were designed first to create a consistent layout.
- ❖ **Backend Routing**: Routes were defined with GET/POST methods, form handling, and Flask sessions.
- ❖ **Database Integration**: SQLAlchemy ORM was used for database abstraction, ensuring easy querying and relational management.
- ❖ **URL Shortening**: Implemented using pyshorteners, which abstracts API calls to TinyURL.
- ❖ **History Management**: URLHistory table keeps records per user, and queries are sorted by timestamp descending.
- ❖ **Flashes and Feedback**: Flash messages were implemented across all forms for user feedback.

The combination of Flask + SQLAlchemy + Bootstrap + pyshorteners provides a full-stack solution for a clean, functional URL shortener web application.

6. **Conclusion**: This project demonstrates the complete lifecycle of a web application: design, backend development, database integration, and frontend styling.

- ❖ Users can sign up, log in, shorten URLs, view history, and log out securely.
- ❖ The application enforces data validation, session management, and user-specific history.
- ❖ Proper styling ensures a responsive, user-friendly interface.
- ❖ Key challenges were resolved, including database errors, session handling, and UI alignment issues.

Overall, the project successfully implements all the intended functionality and is ready for deployment or submission.

Images :

Welcome to URL Shortener

Tired of long URLs? Our app shortens them for you. Copy, save, and share shortened links easily. Log in or sign up to manage your links.

[Login](#)[Sign Up](#)

Created by Souryadip Mallick

Home Page

Login

Username

Password

[Login](#)[Go back to Home](#)

Don't have an account? [Sign Up](#)

Login Page

Sign Up

Username

Password

[Sign Up](#)[Go back to Home](#)

Already have an account? [Login](#)

Sign Up Page

Welcome, [m.souryadip](#) [History](#) [Logout](#)

URL Shortener

Shorten URL

Shortened URL will appear here [Copy](#)

Dashboard Page

Your URL History [Back to Dashboard](#)

#	Original URL	Shortened URL	Timestamp
1	https://bigcatsindia.com/royal-bengal-tiger-information-and-facts/	https://tinyurl.com/2bh3to9r	2026-01-11 01:56:49
2	https://www.whatsapp.com	https://tinyurl.com/p6kortb	2026-01-11 01:35:53
3	https://online.innomatics.in/v3/myaccount/course/189669/lesson/265846?lesson=265846&section=449015&class_id=436588https://online.innomatics.in/v3/myaccount/course/189669/lesson/265846?lesson=265846&section=449015&class_id=436588	https://tinyurl.com/2bsxv5bd	2026-01-11 01:33:47

History Page

You have been logged out. [×](#)

Login

Username

Password

Login

[Go back to Home](#)

Don't have an account? [Sign Up](#)

After Log out